

Pragmatic Emoji Prediction Using Twitter-Trained DistilBERT Model

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Abstract

Emojis have become a vital form of communication - so much so that systems have begun suggesting them in place of words. Emoji prediction systems in modern messaging platforms overwhelmingly rely on literal keyword matching however, failing to capture the pragmatic, cultural, and socially nuanced ways people actually use emojis in everyday communication. This work addresses that gap by introducing a new methodology for building a context-aware model capable of interpreting and predicting three useful emojis that match the tone, stance, and indirect meaning of a given sentence. Drawing on linguistic theories of pragmatic inference and leveraging a Transformer-based architecture fine-tuned for multi-label classification, the system learns to associate textual cues with the non-literal emoji choices common in natural conversation. The results demonstrate that the model captures pragmatic emoji use more effectively than literal baselines, suggesting that emoji prediction can be significantly improved by modeling social meaning rather than surface semantics. This work contributes both a practical application and a conceptual re-framing of emoji prediction as a problem of pragmatic language understanding rather than simple lexical substitution.

1 Introduction

Emojis have become an integral part of everyday digital communication, functioning not merely as decorative symbols but as meaningful cues that shape tone, emotion, and social intent. Results show that approximately 92% of online communications use emojis daily, including text, email, and social media [2]. In addition to conveying meaning, the usage of emojis affects the perception of the receiver as well. The use of emojis in text messages improves perceived responsiveness of the sender, directly influencing likability or closeness of them as well [7]. Despite their ubiquity, most

emoji-suggestion systems continue to rely on literal keyword matching or surface-level semantic associations. Current predictive systems fail to account for these nuances because they treat emojis as direct lexical substitutes rather than as pragmatic markers embedded in social interaction. Capturing these functions of prosody requires models that can interpret text beyond its surface form and recognize the subtle cues that guide non-literal emoji choice.

Considering the large importance emojis play in the world, especially in a world based on quick and rapid communication, we looked towards prevalent domains where text is converted into emojis. The first and most obvious result was Apple's iMessage. It can quickly be recognized that iMessages emoji suggestions are far too literal, however. When a user typed "I'm dead", for example, conventional systems tend to recommend an emoji with X's over its eyes, although contemporary usage overwhelmingly pairs the phrase with a laughing emoji to signal amusement rather than mortality (figure 1). This gap between literal interpretation and pragmatic meaning reflects a broader challenge: emoji use is deeply contextual, culturally shaped, and often intentionally indirect. Today, many emojis are used in contexts outside of their direct meaning. This was the driving inspiration for this work and how a more accurate system could be designed. Additional implementations of this project can be found, though most examples support the prediction of one out of twenty emojis - many of which did not exist within the top 100 most frequently used emojis discovered in this approach.

This work addresses that challenge by reframing emoji prediction as a problem of pragmatic language understanding. Using a Transformer-based architecture fine-tuned for multi-label classification, the system learns to associate textual clues with the culturally shaped and non-literal emoji patterns common in natural conversation. By leveraging contextual embeddings, multihot label en-

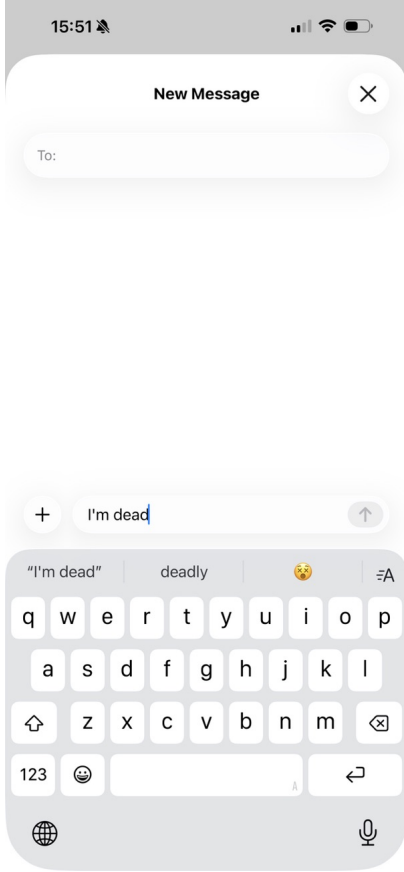


Figure 1: iMessage Emoji Suggestion

coding, and a training strategy designed to preserve pretrained linguistic knowledge, the model moves beyond keyword matching and toward a more human-like interpretation of emoji use. The goal of this research is twofold: to develop a practical, context-aware emoji prediction system and to demonstrate that modeling pragmatic meaning leads to more accurate and socially aligned emoji suggestions. In doing so, this work contributes to both computational linguistics and user-facing communication tools, highlighting the importance of pragmatics in modern language technologies.

2 Background and Related Work

Emoji prediction has emerged as a useful task for studying expressive meaning in digital communication, and prior work has explored various approaches to modeling this. This section reviews relevant research on emoji prediction, discusses existing datasets, and situates the present work within this landscape.

2.1 Emoji Prediction with Transformer Models

Recent advances in natural language processing have led to the adoption of Transformer-based architectures for emoji prediction tasks. One notable example is the work presented by user Harsh CO_195 in their notebook *BERT Emoji Prediction* [6], which fine-tuned a BERT [4] model to classify tweets into one of twenty emojis. Their approach demonstrated that contextual embeddings substantially outperform traditional bag-of-words or keyword-matching methods. Using the dataset introduced by user harias [5], which contains tweets labeled with a limited set of twenty emojis, their model achieved a weighted F1-score of 0.92 and a macro F1-score of 0.88 between the six most common emojis. These results highlight the effectiveness of Transformer models for capturing semantic and syntactic cues associated with emoji usage.

However, the dataset used in the notebook created by Harsh CO_195 includes several emojis that are relatively uncommon in everyday communication, such as the Christmas tree emoji or the American flag emoji. Moreover, their evaluation focused on the six most frequent emojis, which may inflate performance metrics due to class imbalance. While their work demonstrates the feasibility of BERT-based emoji prediction, it does not address the broader challenge of modeling pragmatic or socially nuanced emoji use.

2.2 Limitations of Existing Datasets

An issue mentioned previously with harias’s dataset, most publicly available emoji prediction datasets are constrained by limited label sets and do not reflect the full diversity of emoji usage in natural conversation. Additionally, many datasets include only a single emoji label per tweet, despite the fact that users frequently employ multiple emojis to convey tone, emphasis, or layered meaning. These limitations motivate the need for larger, more representative datasets and modeling approaches that account for multi-label, context-dependent emoji usage.

Building upon this, many datasets reference tweet ID’s rather than providing raw text, necessitating the use of X’s paid API to collect data. This can be seen by Capallo’s Twemoji dataset on Kaggle [1]. While this may provide more up-to-date tweets and emoji use, usage can get expensive

quickly.

2.3 Building on Prior Work

The present work builds on the methodological foundation established in the *BERT Twitter Prediction* notebook by leveraging a similar architecture known as DistilBERT [10] and fine-tuning it for emoji prediction. However, it differs from prior work in several key ways:

- **Expanded Label Space.** Instead of restricting prediction to twenty emojis, this work uses a dataset containing one hundred emojis, providing a more realistic and diverse representation of emoji usage.
- **Multi-Label Prediction.** Unlike prior single-label approaches, this model predicts *three* emojis per input, reflecting the common practice of using multiple emojis to express tone, stance, or pragmatic nuance.
- **Pragmatic Focus.** While previous work emphasizes literal or sentiment-based emoji prediction, this project explicitly targets pragmatic meaning and culturally shaped uses on how emojis function as social and expressive cues rather than direct semantic substitutes.
- **Root-Form Normalization.** This work introduces a normalization procedure that collapses emoji variants (e.g., skin-tone modifiers) into root forms, reducing sparsity and improving generalization.
- **Human Evaluation.** In addition to quantitative metrics, this project incorporates human judgments to assess whether predicted emojis align with intuitive, context-driven interpretations.

Together, these contributions extend prior research by reframing emoji prediction as a problem of pragmatic language understanding rather than simple classification. Furthermore, this work demonstrates that Transformer models can capture aspects of indirect meaning and social expression that are overlooked by literal or frequency-based baselines.

3 Data

A central challenge in developing a context-aware emoji prediction model is obtaining a dataset that reflects natural, pragmatic emoji use. Because direct access to private

messaging data is not feasible, public social media platforms offer the closest approximation to real conversational language. Twitter (now X) is particularly well-suited for this purpose due to its informal tone, high emoji frequency, and widespread use across diverse communities. However, recent changes to the platform’s API (specifically the introduction of a paywall) make large-scale data collection prohibitively difficult without paid access. As a result, this work relies on publicly available datasets hosted on Kaggle.

Dataset 1: Raw Tweets (*EmojiifyData-EN*)

The primary dataset used in this study is *EmojiifyData-EN* [8], a large-scale collection of raw tweets gathered in 2018 prior to the API restrictions. This dataset contains over 18 million rows of text, each including at least one emoji. Because the tweets are unprocessed and reflect natural user behavior, they provide a rich source of emoji usage patterns.

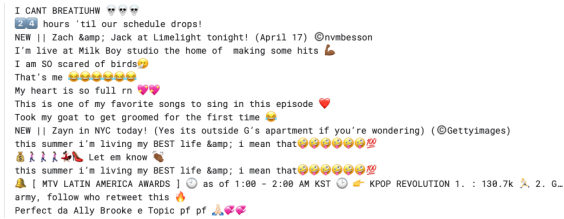


Figure 2: Sample from the *EmojiifyData-EN* raw tweet dataset

Dataset 2: Emoji Mapping (*Tweet2Emoji*) To support emoji extraction and normalization, the *Tweet2Emoji* dataset [9] was used as a reference for Unicode emoji characters. This dataset contains 1,791 emoji symbols and serves as a comprehensive mapping resource, ensuring that all emojis present in the *EmojiifyData-EN* corpus can be accurately identified and processed.

3.1 Data Preprocessing

To prepare the raw tweet corpus for multi-label emoji prediction, preprocessing focused primarily on extracting emojis, normalizing their variants, and filtering the dataset to match the goals of this study. Because the *EmojiifyData-EN* dataset consists of unprocessed social media text, the emphasis was not on traditional text cleaning but rather on accurately identifying emojis, reducing them to consistent root forms, and structuring the data so that each tweet could be represented as a set of unique emoji labels.

Δ	Δ category	Δ title	Δ shorts
1791 unique values	people symbols Other (1035)	27% 15% 58%	1448 unique values
🤔	people	grinning face	[grinning]
😊	people	grinning face with smiling eyes	[grin]
😂	people	face with tears of joy	[joy]
😄	people	smiling face with open mouth	[smiley]
😁	people	smiling face with open mouth and smiling eyes	[smile]

Figure 3: Sample from *Tweet2Emoji* emoji mapping

3.1.1 Emoji Extraction and Normalization

	title	unicode_short	unicode
150	raised hand	270b	👉
151	raised hand type	270b_1f3fb	👉🏻
152	raised hand type	270b_1f3fc	👉🏼
153	raised hand type	270b_1f3fd	👉🏽
154	raised hand type	270b_1f3fe	👉🏾
155	raised hand type	270b_1f3ff	👉🏿

Figure 4: Example of variants of the raised hand emoji

Many emojis contain a wide variety of variants. These variants are most commonly used to modify the skin color of the various human-based emojis, although they can be seen in other types of emojis such as flags or hearts. As can be seen in figure 4, we consider the raised hand emoji. Paying close attention to the “unicode_short” column, we notice that every colored variant is a compound of the root yellow emoji. If we take this emoji and split its hex code at the underscore, we receive the root yellow emoji and a secondary emoji, which is known as a *modifier emoji* (which in this case represents the skin color). This can be seen in figure 5.

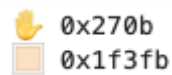


Figure 5: Example of variants of the *raised hand* emoji

These emojis, though visually different, functionally represent the same intent. We then began preparing the master dataframe. Using the *Tweet2Emoji* unicode mapping, each tweet was scanned for all emoji characters. Because tweets may contain multiple emojis used for emphasis, tone, or layered meaning, each tweet was assigned a *set* of unique emojis that appeared in the tweet rather than a single label. Each extracted emoji in

this set was reduced to its root form as mentioned previously.

3.1.2 Filtering and Quality Control

Several filtering steps were applied to improve dataset quality:

- Tweets containing repeated copies of the same emoji were treated as a single label.
- Tweets containing less than or greater than three emojis were discarded to target the optimal prediction of three emojis.
- Reduced emoji list to the top 100 most frequent emojis (figure 6), discarding any tweets that contained an emoji outside of this subset.

These steps reduced noise while preserving the natural variability of emoji-rich communication.

3.1.3 Label Encoding

Because each tweet may correspond to multiple emojis, labels were encoded using multihot vectors, where each dimension corresponds to a unique emoji in the dataset. A value of 1 indicates the presence of that emoji in the tweet, and 0 indicates its absence. This representation supports multi-label classification and allows the model to predict multiple emojis simultaneously. After all modifications, the dataframe was reduced from over 18 million rows to just below 433,000 rows. The final dataframe can be viewed in figure 7.

3.1.4 Train-Validation Split

The cleaned and encoded dataset was split into training and validation sets using an 80/10/10 ratio. Stratification was applied at the label level to maintain a balanced distribution of emojis across splits, ensuring that both common and rare emojis were represented during evaluation.

4 Model and Approach (Method)

Can a context-aware Transformer model predict the three emojis that best reflect the pragmatic meaning of a given sentence? To answer this question, we defined the task as a multi-label classification problem.

DistilBERT For the model, we chose the pre-trained model *distilbert-base-uncased*. This is a light-weight model with 67M parameters, 6 layers, and 12 bidirectional attention heads. For the model class, we went with *AutoModelForSequenceClassification*. This is a classification head built to run

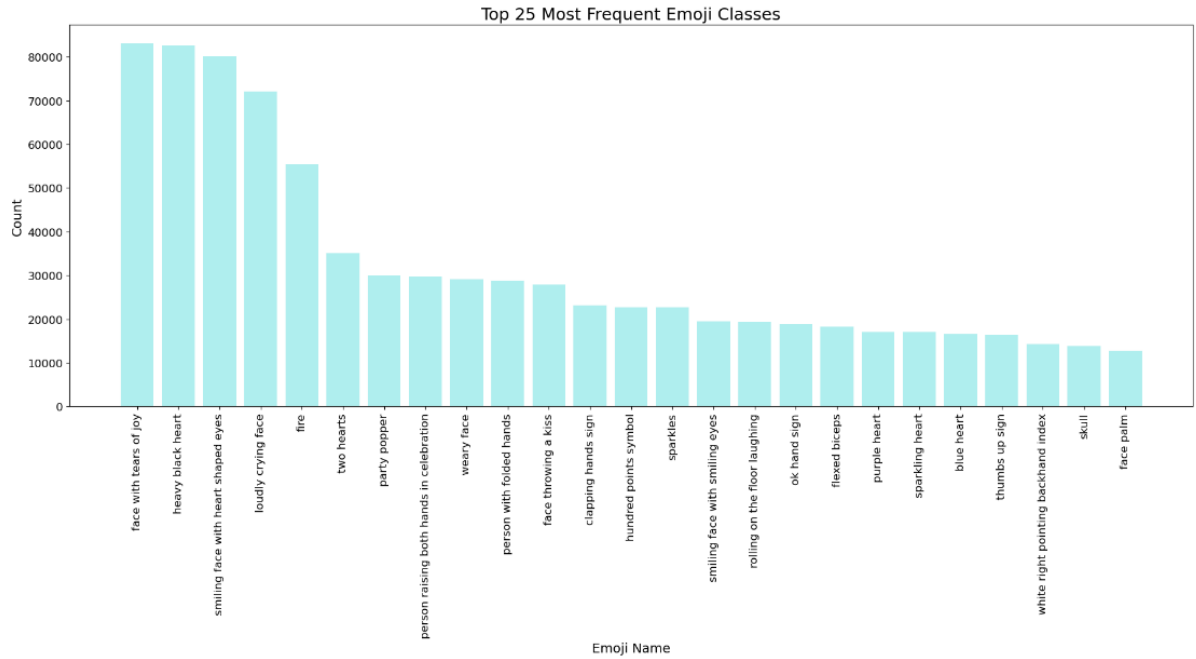


Figure 6: The top 100 most frequent emojis by name

	raw_text	extracted	emojis	encoding
47	If she hating, tell that bitch to let it goooo...	If she hating, tell that bitch to let it goooo...	[👉, 🗿, 🍷]	[0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 1.0, ...]
48	Rihanna looks so good in a dark lip 🍷💋💋	Rihanna looks so good in a dark lip 🍷💋💋	[🍷, 🍷, 🍷]	[0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, ...]
66	that looks so good 🍷🍷🍷🍷🍷	that looks so good 🍷🍷🍷🍷🍷	[🍷, 🍷, 🍷]	[0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, ...]
247	We literally drive eachother crazy...but only c...	We literally drive eachother crazy...but only c...	[👉, 🍷, 🍷]	[0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, ...]
283	I don't care who drops what or when! We're cl...	I don't care who drops what or when! We're cl...	[🍷, 🍷, 🍷]	[0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, ...]
...	...	...	...	...
18883363	Link LSKprice :31.10 Binance is TEMPORARILY...	Link LSKprice :31.10 Binance is TEMPORARILY...	[👉, 🗿, 🍷]	[0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, ...]
18883398	Happy Birthday! Fab boots 🍷🍷🍷	Happy Birthday! Fab boots 🍷🍷🍷	[🍷, 🍷, 🍷]	[0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, ...]
18883455	Primero lo primero, estampa oficial... FIRST ...	Primero lo primero, estampa oficial... FIRST ...	[🍷, 🍷, 🍷]	[0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, ...]
18883485	If my man got me this 🍷💋💋 "Drops hint	If my man got me this 🍷💋💋 "Drops hint	[🍷, 🍷, 🍷]	[0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, ...]
18883573	∓ to think I was going to split my tax ref...	∓ to think I was going to split my tax ref...	[🍷, 🍷, 🍷]	[0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, ...]

Figure 7: The final dataframe after all modifications

on top of DistilBERT and defines the task for the model, which in this case is converting text into a multi-label classification.

4.1 Parameters

- **Epochs: 10**
- **Batch size: 16**
- **Max sequence length: 128**
- **Tokenizer: AutoTokenizer** Set to work with the AutoModelForSequenceClassification model class.
- **Optimizer: AdamW** This optimizer is good for fine-tuning transformer models.
- **Learning Rate: 2e-5** This served as a good middle ground between the model being too strict and not strict enough.
- **Loss Function: BCEWithLogitsLoss** This was created by implementing a tensor with an

inverse frequency function applied to all of the class frequencies. This ensured that more rare emojis had higher weights while common emojis had lower weights.

4.2 Loss Function

Because emoji prediction is a multi-label task, each emoji is treated as an independent binary decision. For this reason, the model uses *BCEWithLogitsLoss*, which applies a sigmoid activation to each output dimension and computes a separate loss for every emoji. This allows the model to assign multiple emojis to a single input rather than forcing a single categorical choice.

To address the strong class imbalance in the dataset, we incorporate inverse-frequency class weights. Emojis that appear less frequently receive higher weights, increasing their contribution to the loss and preventing the model from overfitting to the most common emojis. The weights are computed by dividing the total number of samples by the frequency of each emoji and then normalizing so the average weight is 1. This ensures stable optimization while encouraging the model to learn meaningful representations for rare emojis.

4.3 Prediction Generation

At inference time, the model outputs a vector of logits of length 100, corresponding to the predicted relevance of each emoji. These logits are

passed through a sigmoid function to convert them into independent probabilities in the range $[0, 1]$. Because the task requires predicting exactly three emojis, we apply a top- k selection strategy: the three emojis with the highest predicted probabilities are returned as the model’s output. This avoids the need for a fixed probability threshold and ensures consistent multi-emoji predictions that reflect the model’s learned understanding of pragmatic meaning.

With this approach, we hope to demonstrate that emoji prediction can capture pragmatic meaning and move beyond keyword-matching systems.

5 Evaluation

In order to answer our question of whether or not a Transformer model can predict the three most pragmatic emojis for a given sentence, we evaluated performance against two baseline systems and conducted a small-scale human evaluation. Because the task involves predicting three emojis per input, both baselines and model performance were measured in terms of accuracy on the most common emoji as well as multi-label F1 scores.

5.1 Baselines

Random Baseline. The random baseline selects one emoji from the full label set of 100 possible emojis. This yields an expected accuracy of 1%, reflecting the sparsity of correct labels under random selection.

Most Common Emoji Baseline. A stronger baseline predicts the single most frequent emoji in the dataset for every input. In the processed corpus, the “laughing while crying” emoji occurred 83,195 times out of 432,905 tweets, producing a baseline accuracy of 19.21%. Although simplistic, this baseline captures the heavy-tailed distribution of emoji usage and provides a meaningful point of comparison.

F1 Scores The strongest metric measures performance on multi-label classification through micro, macro, and weighted F1 scores. These three scores measure how well the model performs on the dataset as a whole, how well it performs across each emoji, and a weighted measurement of performance based on emoji frequencies, respectively.

5.2 Human Evaluation

To complement quantitative metrics, a small-scale human evaluation was conducted to assess the model’s predictions from a pragmatic perspective. Participants were presented with a curated set of sentences and asked to select the three emojis they believed were most appropriate for each input. Model predictions were then compared to these human selections to gauge alignment with intuitive, context-driven emoji interpretation.

Although informal, this evaluation provided qualitative evidence that the model often produced emoji sets judged as reasonable or contextually appropriate by human annotators. These results support the claim that the model captures aspects of pragmatic meaning that are not reflected in literal or frequency-based baselines.

5.3 Results

The model was evaluated using both baseline comparisons and multi-label classification metrics. Table 1 summarizes the performance of the random baseline, the most common emoji baseline, and the proposed model.

Method	Accuracy	Micro F1	Macro F1	Weighted F1
Random	1.00%	—	—	—
Most Common	19.21%	—	—	—
Proposed Model	85.62%	44.44%	49.24%	40.97%

Table 1: Performance comparison between baselines and the proposed model.

The model achieved an accuracy of 85.62% on the most common emoji, substantially outperforming both the random baseline (1%) and the most common emoji baseline (19.21%). This demonstrates that the model effectively learns high-frequency pragmatic patterns rather than relying on distributional bias alone.

Beyond single-label accuracy, multi-label performance was assessed using micro, macro, and weighted F1 scores. The model achieved a micro F1 score of 44.44%, a macro F1 score of 49.24%, and a weighted F1 score of 40.97%. These results represent a meaningful improvement over random chance and indicate that the model captures a broad range of emoji usage patterns beyond the most frequent labels. The F1 scores per emoji can be seen in figure 8.

These F1 scores per emoji show a mixed story. As for common emojis with a low F1 score, this can be reasoned with the idea that the model is likely

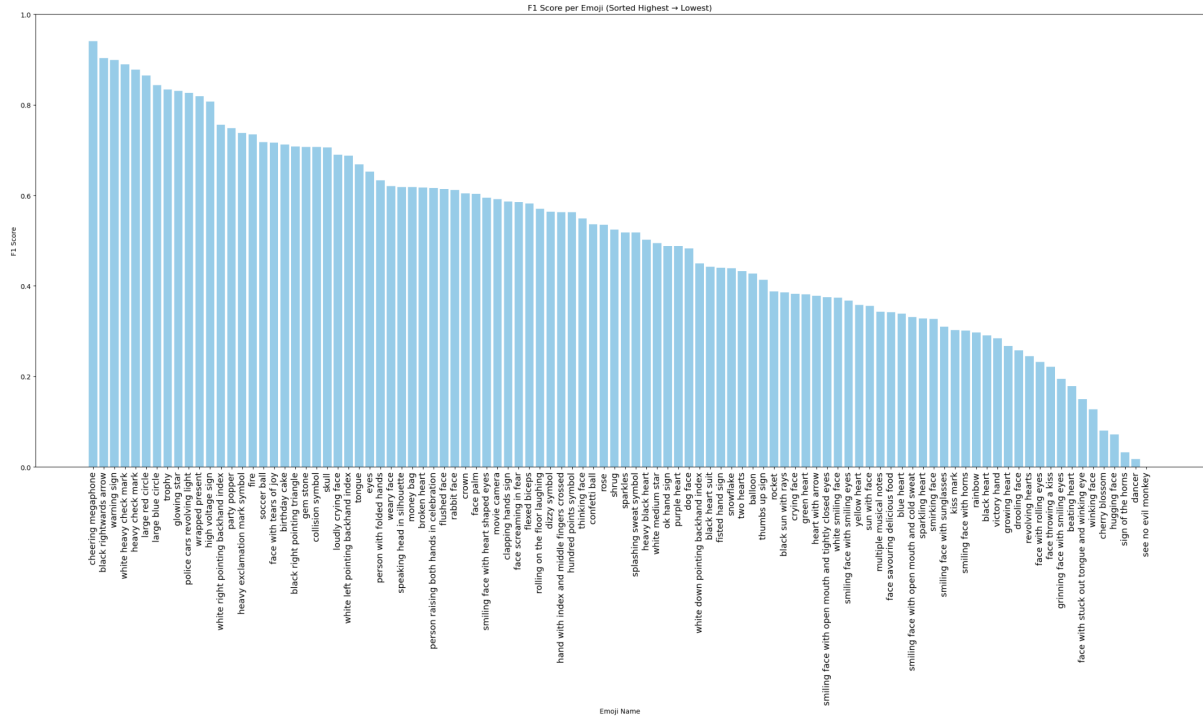


Figure 8: F1 scores per emoji

predicting common emojis too much. Similarly, if a less common emoji has a low F1 score, we can reason that the model is not predicting those emojis enough. The emojis with a higher F1 score tend to be more niche. Consider the example of the birthday cake emoji. This emoji only appears in the context of a birthday and can be easy for the model to accurately predict. If we consider the loudly crying emoji, its meaning can be more nuanced, as it can be used to signify laughing or crying.

The human evaluation shows mixed results, as can be seen in figure 9. For this example, users were asked to guess the three emojis they would expect to appear in a given sentence. Green represents an accurately predicted emoji while orange signifies an incorrect guess that was not included in the models prediction set. These results can be attributed to many factors, including individual differences in demeanor and expression. Some sentences achieved a relatively high score, while more nuanced sentences struggled to produce preferable results.

Overall, the results demonstrate that, while the problem is highly complex, modeling pragmatic meaning, rather than relying on literal or frequency-based heuristics, significantly improves emoji prediction performance. The combination of contextual embeddings and multi-label classification en-

ables the system to better approximate human-like emoji interpretation.

6 Implications

The findings of this work highlight the potential for emoji prediction systems to move beyond literal keyword matching and toward a richer, pragmatically informed understanding of digital communication. The ability to model non-literal meaning has implications not only for messaging platforms, but also for sentiment analysis, conversational agents, and any system that seeks to interpret or generate human-like expressive language.

6.1 Discussion

The model’s performance indicates that pragmatic emoji use can be learned from large-scale social media data, even when labels are noisy and subjective. The substantial improvement over both random and frequency-based baselines shows that contextual cues play a central role in emoji interpretation. However, the moderate F1 scores also reflect the inherent ambiguity of the task: emoji choice is influenced by personal style, cultural background, and conversational context, making it difficult to define a single “correct” label set.

Human evaluation further supports this interpretation. Participants often judged the model’s

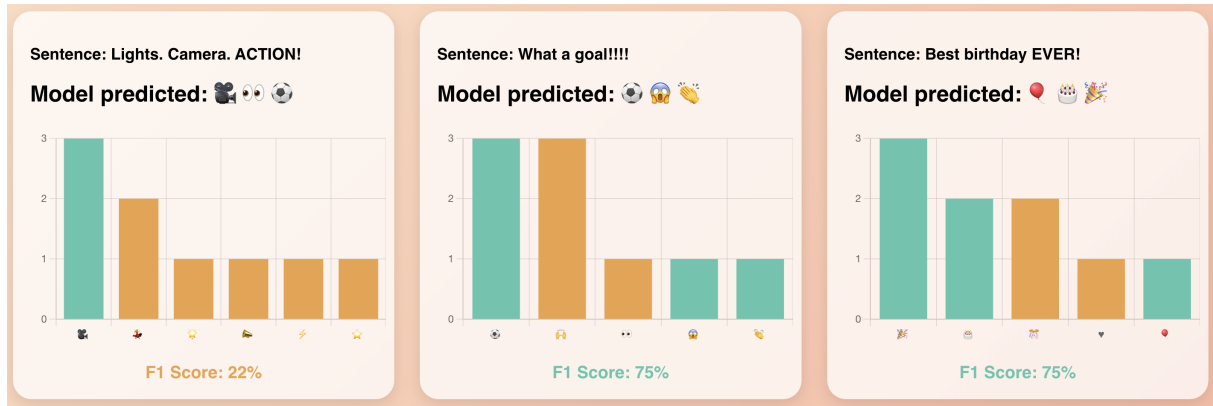


Figure 9: User expectations against models predictions

predictions as reasonable even when they did not match the dataset labels exactly. This suggests that strict accuracy metrics may underrepresent the model’s true pragmatic performance. Future evaluation frameworks may benefit from incorporating semantic similarity or human-centered scoring to better capture appropriateness rather than exact matching.

6.2 Directions for Future Work

Several directions could extend and strengthen this research:

- **Enhanced Text Cleaning.** Incorporating pre-processing steps such as removing hashtags, user mentions, URLs, and boilerplate text could reduce noise and improve the clarity of linguistic cues.
- **Spam and Bot Filtering.** A significant portion of Twitter data is generated by automated accounts. Filtering these out would yield more human-like emoji usage patterns.
- **Emoji Similarity Metrics.** Future models could incorporate semantic or embedding-based similarity between emojis, allowing for graded evaluation rather than strict correctness.
- **Updated or Expanded Datasets.** The dataset predates major shifts in emoji usage. More recent or domain-specific corpora could improve generalizability. This can be seen in *Emoji Frequency* [3], placing importance on up-to-date datasets.
- **Larger-Scale Human Studies.** Broader human evaluation would provide more reliable insight into how well the model aligns with real-world intuition.

- **Fine-tuned Emoji Thresholds.** Individual thresholds could be applied to each emoji to ensure the fair balance of classes. More common emojis would have a higher threshold and less common emojis would have lower thresholds.

7 Conclusion

This work set out to address a fundamental limitation in current emoji-suggestion systems: their reliance on literal, surface-level associations that fail to reflect how people actually use emojis in everyday communication. By reframing emoji prediction as a problem of pragmatic language understanding rather than lexical substitution, this research demonstrates that models can learn to capture the cultural, contextual, and socially meaningful patterns that shape non-literal emoji use.

Beyond improving emoji suggestion, this work contributes to a broader understanding of how language technologies can better account for the subtleties of digital communication. As emojis continue to evolve as a form of expressive, culturally embedded language, systems that model their pragmatic use will be essential for building more intuitive and socially aware communication tools. Future work may extend this approach to multi-modal inputs, cross-cultural datasets, or real-time messaging environments, further bridging the gap between human communication practices and the models designed to support them.

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